# Northwind Traders Dashboard

# NORTHWIND TRADERS DASHBOARD

**Data Analysis and Business Intelligence Project**

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# ABSTRACT

The Northwind Traders Dashboard project is developed using SQL, Microsoft Excel, Power BI, Power Query, and DAX to analyze and visualize business data effectively. The project focuses on transforming raw organizational data into meaningful business insights through interactive dashboards and visual reports.

The dashboard analyzes sales performance, customer behavior, employee productivity, supplier distribution, product demand, and shipping operations. Data cleaning and preprocessing techniques were applied to improve data quality and consistency before performing analysis.

Interactive visualizations such as KPI cards, bar charts, line charts, pie charts, maps, and slicers were used to represent business metrics effectively. The project helps organizations monitor performance, identify trends, and support data-driven decision-making.

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# CHAPTER I

## 1.1 Introduction

The Northwind Traders Dashboard project is a Business Intelligence and Data Analytics project developed to analyze organizational data and generate meaningful business insights. The project focuses on sales analysis, customer behavior, employee performance, supplier distribution, and product demand using modern Business Intelligence tools.

The Northwind Traders dataset contains information related to customers, products, employees, suppliers, orders, shipping operations, and product categories. These interconnected datasets provide a real-world business scenario for performing analytical operations and dashboard development.

The project uses SQL for data querying, Microsoft Excel for Exploratory Data Analysis (EDA), Power BI for dashboard creation, Power Query for data transformation, and DAX for advanced calculations and KPI generation.

The dashboard contains interactive visualizations that help users identify trends, compare performance metrics, and make strategic business decisions efficiently.

## 1.2 Objectives

- To analyze company sales performance
- To identify top-selling products
- To analyze customer purchasing behavior
- To evaluate employee productivity
- To visualize regional sales distribution
- To create interactive dashboards using Power BI
- To support data-driven business decisions

### 1.3 Problem Statement

The objective of this project is to create an interactive Power BI dashboard that provides meaningful business insights related to sales, customers, employees, products, and suppliers.

The dashboard helps stakeholders:

- Monitor business performance
- Identify sales trends
- Analyze customer behavior
- Improve operational efficiency
- Make strategic business decisions

# CHAPTER II

## DATASET DESCRIPTION :-

The dataset used in this project is the Northwind Traders dataset, which contains sales and operational information for a fictional company specializing in food import and export business.

### Tables Used

**Customers Table**

Stores customer information such as company name, city, region, and country.

**Employees Table**

Contains employee details including employee names, job titles, and hire dates.

**Orders Table**

Stores order details such as order date, shipping information, and freight.

**Order Details Table**

Contains product quantity, unit price, and discount information.

**Products Table**

Stores product names, pricing, categories, and stock levels.

**Suppliers Table**

Contains supplier company information and regions.

**Categories Table**

Stores product category details.

**Shippers Table**

Contains shipping company information.

# CHAPTER III

## LITERATURE SURVEY

Business Intelligence systems play an important role in modern organizations by converting raw business data into useful insights. Several studies emphasize the importance of data visualization, dashboard analytics, and interactive reporting in improving organizational decision-making.

Power BI dashboards are widely used because they provide:

- Interactive reporting
- Real-time visualization
- Dynamic filtering
- KPI monitoring
- Data-driven insights

Research studies also highlight the importance of:

- Data cleaning
- Data preprocessing
- SQL analysis
- Exploratory Data Analysis (EDA)

These techniques improve the accuracy and reliability of analytical reports.

# CHAPTER IV

## SYSTEM MODELLING

### 4.1 System Architecture

The project consists of the following components:

**Data Collection**

Data is collected from the Northwind Traders dataset.

**Data Cleaning**

Data preprocessing is performed using Power Query and Excel.

**SQL Analysis**

SQL queries are used to analyze customer orders, product sales, and employee performance.

**Dashboard Development**

Interactive dashboards are developed using Power BI.

**Visualization**

Charts, KPI cards, maps, slicers, and filters are used for data representation.

### 4.2 Technology Stack

| Technology | Purpose |
|---|---|
| SQL | Data querying |
| Excel | EDA and preprocessing |
| Power BI | Dashboard creation |
| Power Query | Data transformation |
| DAX | KPI calculations |

# CHAPTER V

## DATA CLEANING AND PREPROCESSING

Data cleaning is an important step in the project because raw datasets often contain missing values, duplicate records, and inconsistent data.

**Steps Performed**

- Removed duplicate records
- Handled missing values
- Corrected data types
- Renamed columns
- Created relationships between tables
- Removed unnecessary columns

Power Query was used extensively for:

- Data transformation
- Filtering
- Splitting columns
- Merging tables

The cleaned dataset improved the accuracy and reliability of the analysis.

# CHAPTER VI

**SQL ANALYSIS**

SQL queries were used to analyze business performance and generate meaningful insights.

**Example SQL Queries**

**Top Selling Products**

SELECT ProductName, SUM(Quantity) AS TotalSales

FROM OrderDetails

GROUP BY ProductName

ORDER BY Total Sales DESC;

**Customer Orders**

SELECT Customer ID, COUNT(Order ID) AS Total Orders

FROM Orders

GROUP BY CustomerID;

**Employee Performance**

SELECT Employee ID, COUNT(Order ID) AS Orders Handled

FROM Orders

GROUP BY EmployeeID;

# CHAPTER VII

## EXPLORATORY DATA ANALYSIS (EDA)

EDA was performed using Microsoft Excel.

**Analysis Performed**

- Monthly sales trends
- Product category analysis
- Customer purchasing behavior
- Employee performance analysis
- Regional sales analysis

**Visualization Techniques**

- Pivot Tables
- Bar Charts
- Pie Charts
- Line Charts
- Conditional Formatting

EDA helped identify trends and business patterns effectively.

# CHAPTER VIII

## POWER BI DASHBOARD DEVELOPMENT

The Power BI dashboard was designed to provide interactive and meaningful business insights.

### Dashboard Features

- KPI Cards
- Interactive Filters
- Slicers
- Dynamic Charts
- Navigation Buttons
- Drill-through Reports

### Visualizations Used

| Visualization | Purpose |
|---|---|
| KPI Cards | Display sales and profit |
| Line Chart | Monthly sales trends |
| Bar Chart | Top products |
| Pie Chart | Category analysis |
| Map Chart | Regional sales distribution |
| Tree Map | Product category performance |

# CHAPTER IX

## DAX MEASURES

### Total Sales

Total Sales = SUM(OrderDetails[UnitPrice])

### Total Orders

Total Orders = COUNT(Orders[OrderID])

### Profit Margin

Profit Margin = DIVIDE([Total Profit],[Total Sales])

# CHAPTER X

## RESULTS AND DISCUSSION

The dashboard successfully generated meaningful business insights from the dataset.

### Key Findings

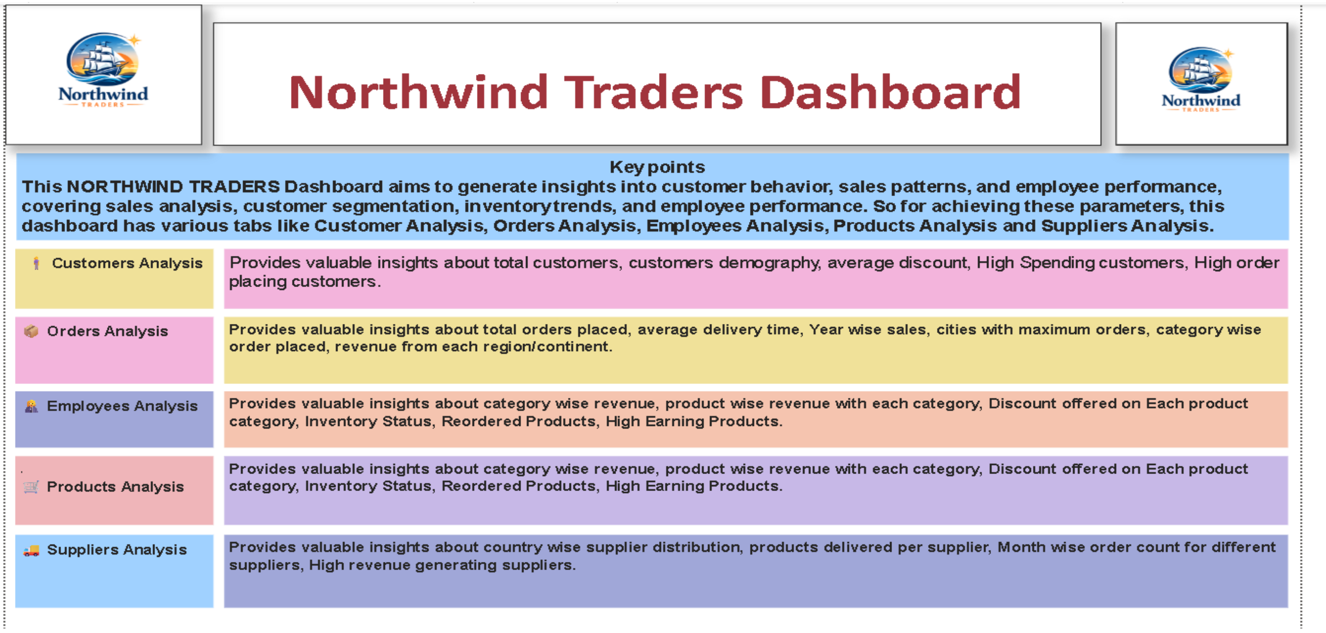
- Beverage category generated the highest revenue
- USA region contributed maximum orders
- Certain products showed seasonal demand patterns
- Repeat customers contributed significantly to revenue
- Employee performance varied across regions

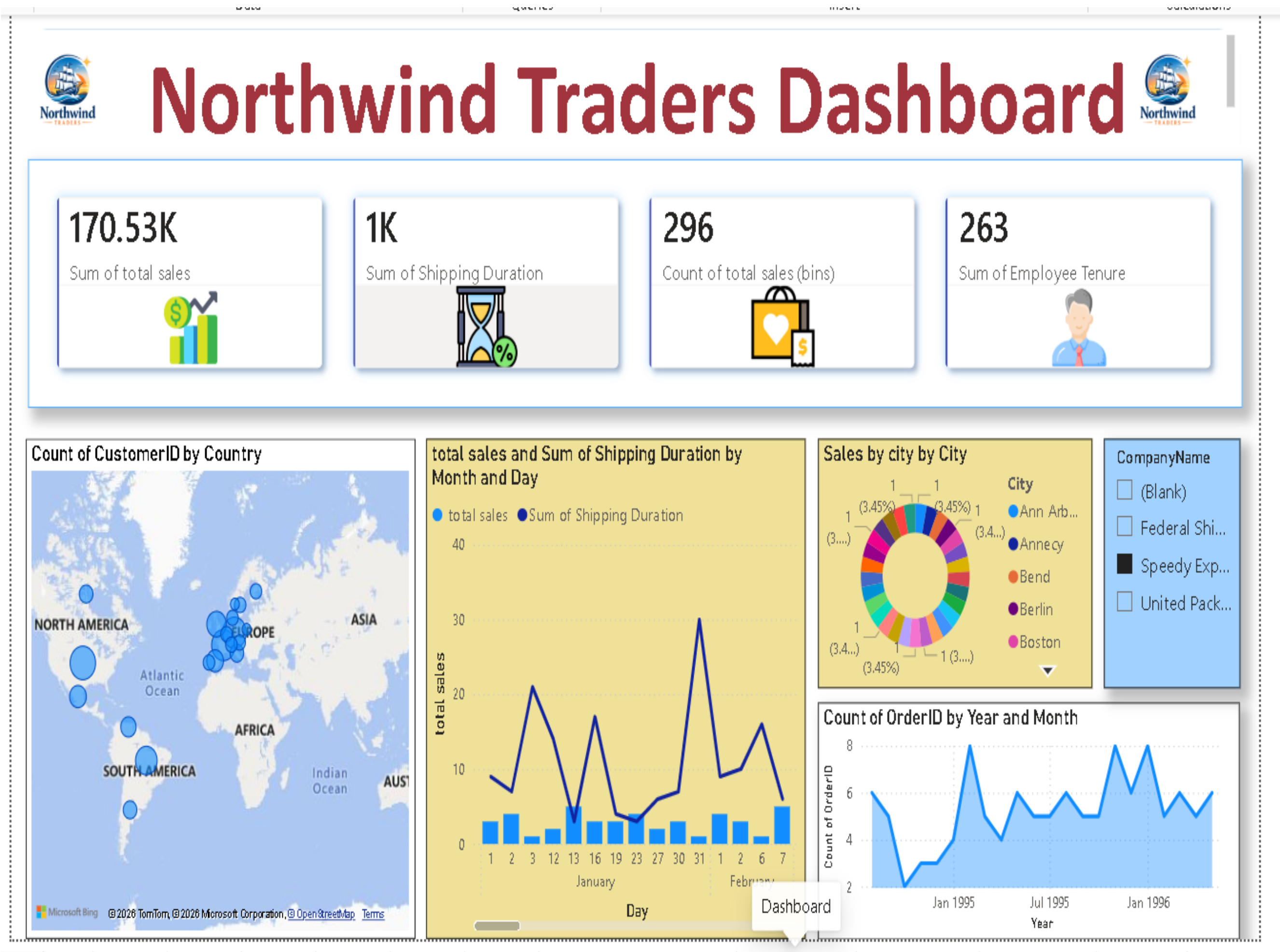
### Dashboard Performance

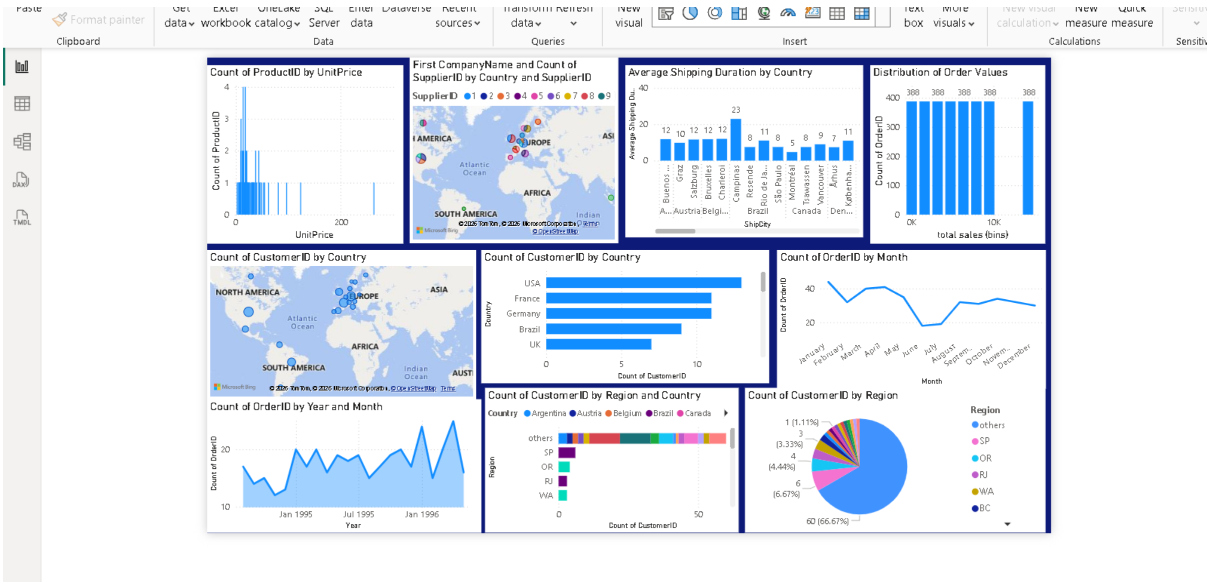
The Power BI dashboard exceeded expectations in:

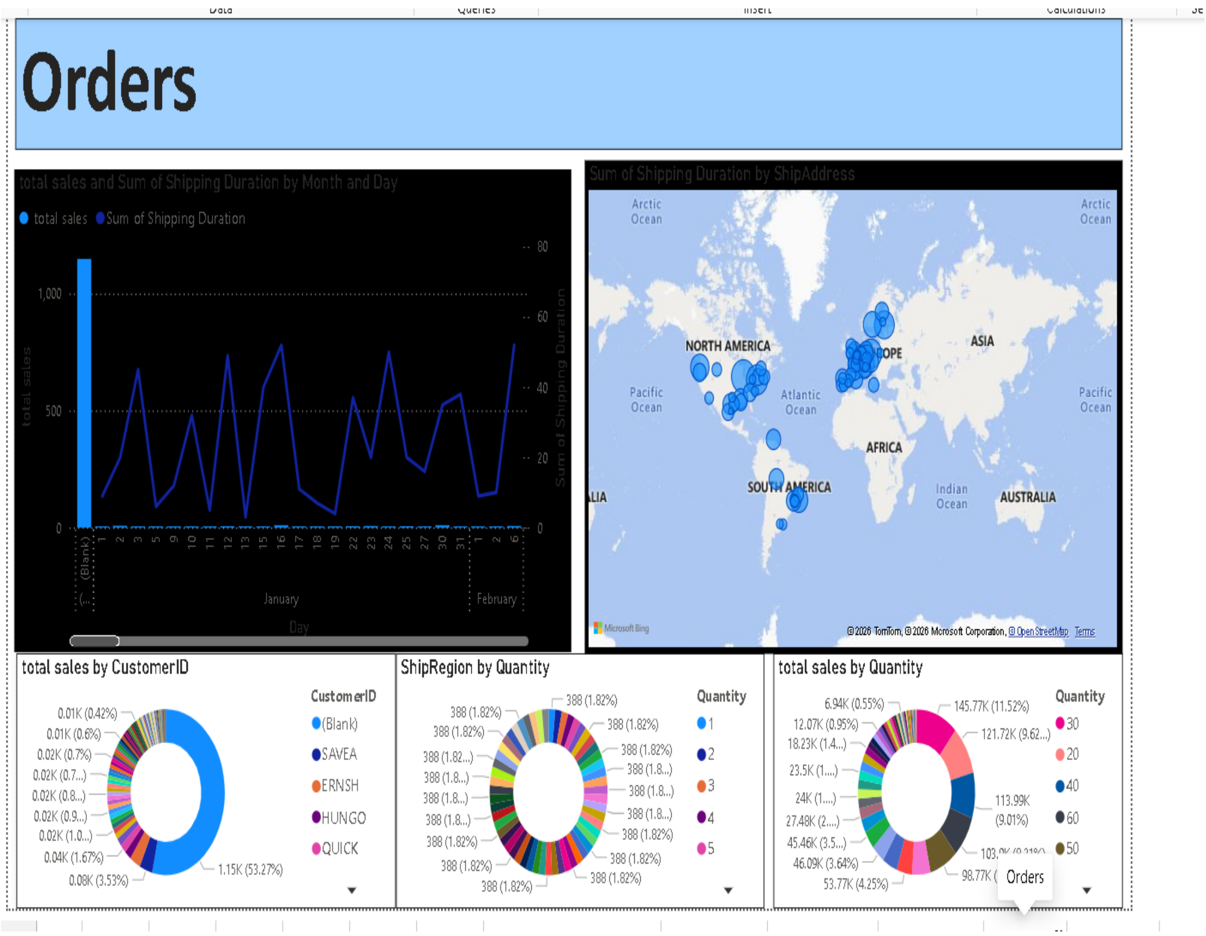
- Visualization quality
- Interactive reporting
- Dashboard design
- Data storytelling
- Insight generation

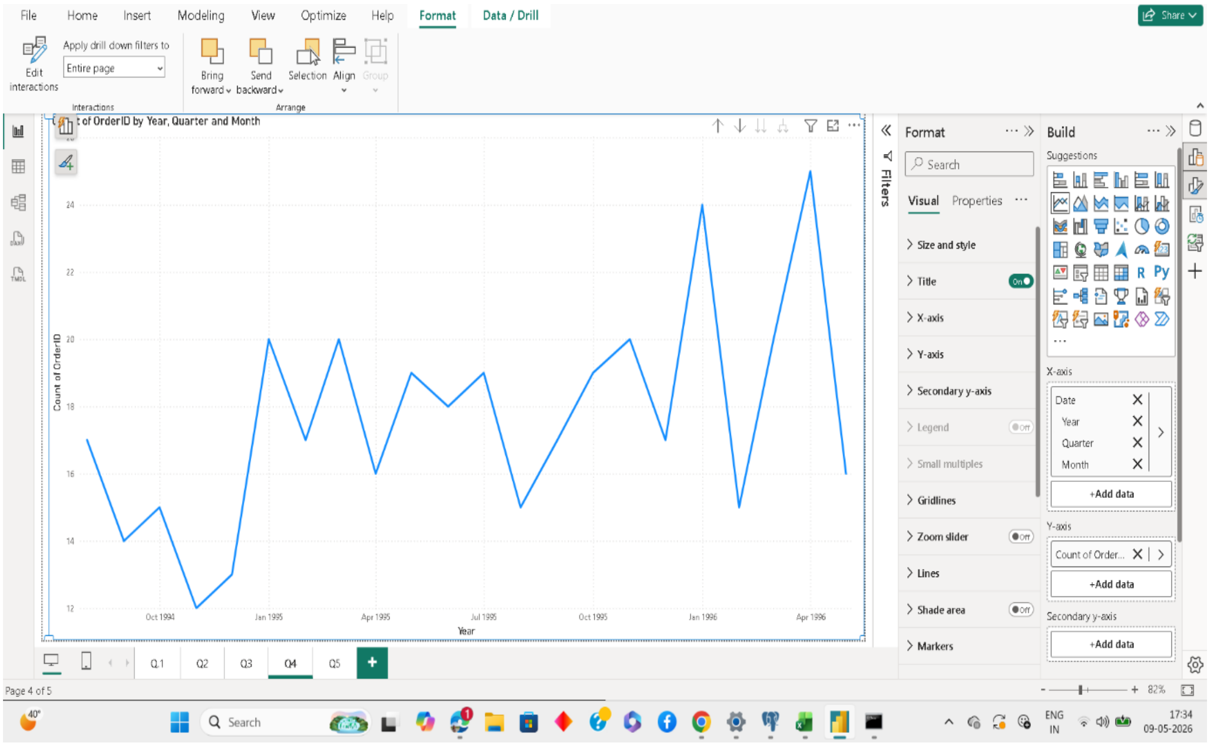
**Output:-**

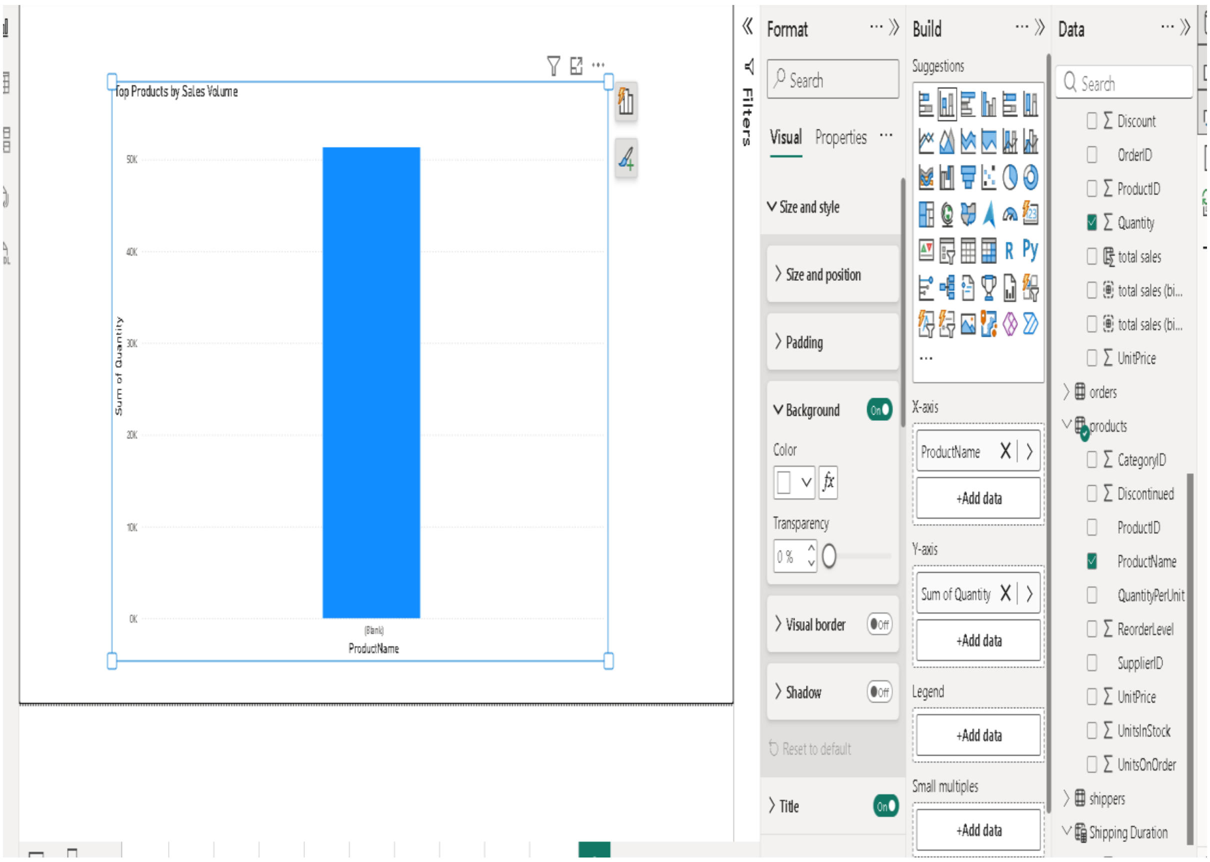

Northwind Traders Dashboard
Key points
This NORTHWIND TRADERS Dashboard aims to generate insights into customer behavior, sales patterns, and employee performance, covering sales analysis, customer segmentation, inventory trends, and employee performance. So for achieving these parameters, this dashboard has various tabs like Customer Analysis, Orders Analysis, Employees Analysis, Products Analysis and Suppliers Analysis.
Customers Analysis
Provides valuable insights about total customers, customers demography, average discount, High Spending customers, High order placing customers.
Orders Analysis
Provides valuable insights about total orders placed, average delivery time, Year wise sales, cities with maximum orders, category wise order placed, revenue from each region/continent.
Employees Analysis
Provides valuable insights about category wise revenue, product wise revenue with each category, Discount offered on Each product category, Inventory Status, Reordered Products, High Earning Products.
Products Analysis
Provides valuable insights about category wise revenue, product wise revenue with each category, Discount offered on Each product category, Inventory Status, Reordered Products, High Earning Products.
Suppliers Analysis
Provides valuable insights about country wise supplier distribution, products delivered per supplier, Month wise order count for different suppliers, High revenue generating suppliers.

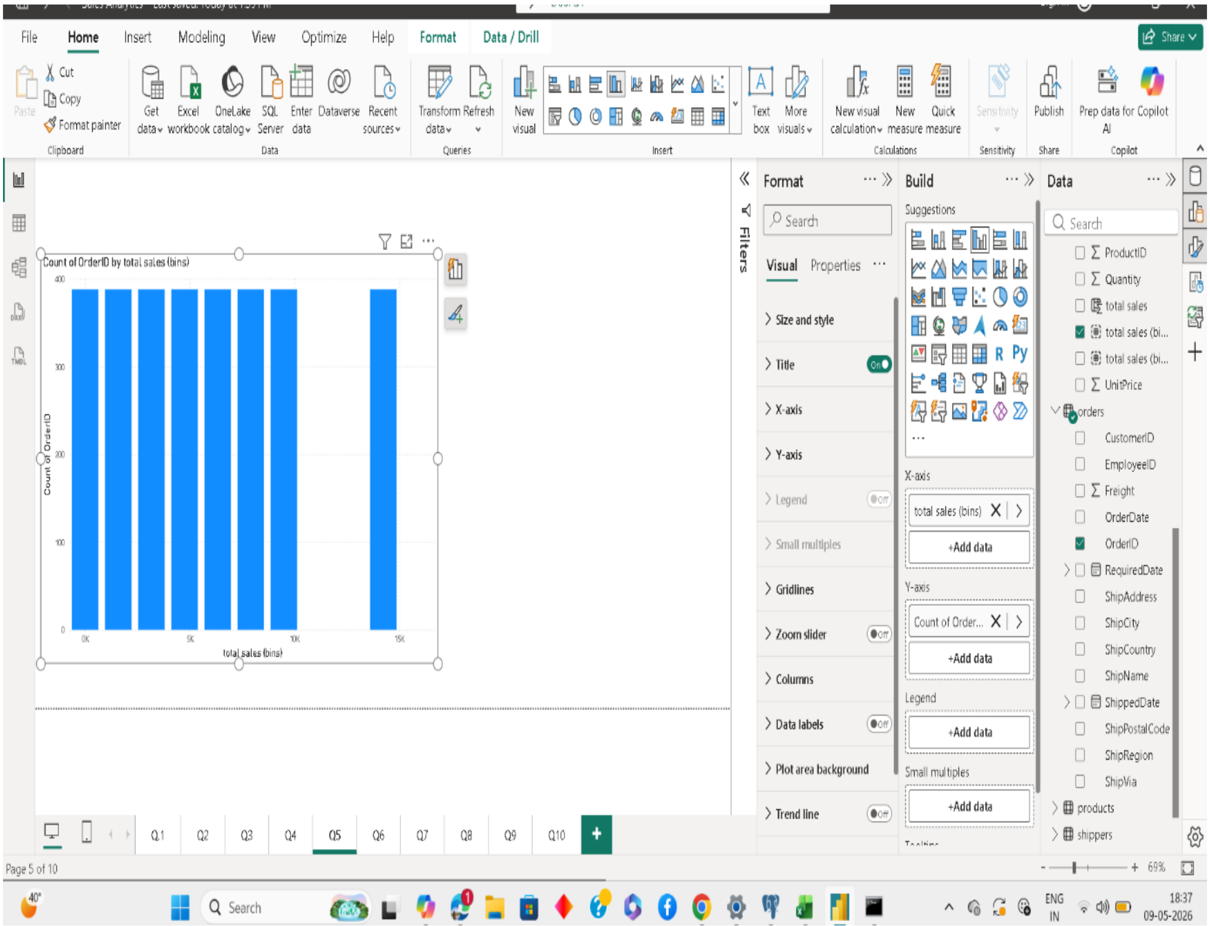

Northwind Traders Dashboard
170.53K
Sum of total sales
1K
Sum of Shipping Duration
296
Count of total sales (bins)
263
Sum of Employee Tenure
Count of CustomerID by Country
NORTH AMERICA
EUROPE
ASIA
Atlantic Ocean
AFRICA
SOUTH AMERICA
Indian Ocean
total sales and Sum of Shipping Duration by Month and Day
total sales
Sum of Shipping Duration
January
Day
Sales by city by City
City
Ann Arb...
Annecy
Bend
Berlin
Boston
CompanyName
(Blank)
Federal Shi...
Speedy Exp...
United Pack...
Count of OrderID by Year and Month
Jan 1995
Jul 1995
Jan 1996
Year
Dashboard


Count of ProductID by UnitPrice
First CompanyName and Count of SupplierID by Country and SupplierID
Average Shipping Duration by Country
Distribution of Order Values
Count of CustomerID by Country
Count of CustomerID by Country
Count of OrderID by Month
Count of OrderID by Year and Month
Count of CustomerID by Region and Country
Count of CustomerID by Region


Orders
total sales and Sum of Shipping Duration by Month and Day
Sum of Shipping Duration by ShipAddress
total sales by CustomerID
ShipRegion by Quantity
total sales by Quantity


Count of OrderID by Year, Quarter and Month
Count of OrderID
Year
Format
Build
Page 4 of 5


Top Products by Sales Volume
Sum of Quantity
ProductName
Format
Build
Data


File
Home
Insert
Modeling
View
Optimize
Help
Format
Data / Drill
Share
Cut
Copy
Format painter
Clipboard
Get data
Excel workbook
OneLake catalog
SQL Server
Enter data
Dataverse
Recent sources
Data
Transform data
Refresh
Queries
New visual
Insert
Text box
More visuals
New visual calculation
New measure
Quick measure
Calculations
Sensitivity
Publish
Share
Prep data for Copilot
AI
Copilot
Count of OrderID by total sales (bins)
400
300
200
100
0
Count of OrderID
0K
5K
10K
15K
total sales (bins)
Filters
Format
Search
Visual
Properties
Size and style
Title
X-axis
Y-axis
Legend
Small multiples
Gridlines
Zoom slider
Columns
Data labels
Plot area background
Trend line
Build
Suggestions
X-axis
total sales (bins)
+Add data
Y-axis
Count of Order...
+Add data
Legend
+Add data
Small multiples
+Add data
Data
Search
ProductID
Quantity
total sales
total sales (bi...
total sales (bi...
UnitPrice
orders
CustomerID
EmployeeID
Freight
OrderDate
OrderID
RequiredDate
ShipAddress
ShipCity
ShipCountry
ShipName
ShippedDate
ShipPostalCode
ShipRegion
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Q5
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Q8
Q9
Q10
Page 5 of 10
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# CHAPTER XI

## CONCLUSION

The Northwind Traders Dashboard project successfully transformed raw organizational data into meaningful visual insights using SQL, Excel, and Power BI.

The project improved understanding of:

- Sales trends
- Customer behavior
- Product performance
- Operational efficiency

The dashboard supports better decision-making by providing interactive reports and visual analytics.

The project also enhanced practical knowledge in:

- SQL querying
- Data cleaning
- Exploratory Data Analysis
- Dashboard development
- Data visualization

# CHAPTER XII

## FUTURE SCOPE

Future improvements can include:

- Real-time data integration
- AI-based forecasting
- Predictive analytics
- Mobile dashboard optimization
- Advanced customer segmentation
- Cloud database integration

# REFERENCES


1. Microsoft Power BI Documentation
2. SQL Documentation
3. Microsoft Excel Documentation
4. Northwind Traders Dataset
5. Microsoft Learn Platform
6. Business Intelligence Research Papers